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A review of spatial localization methodologies for the electric vehicle charging infrastructure

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ABSTRACT

With view to the high share of the transport sector in total energy consumption, e-mobility should play an important role within the transition of the energy systems. Policymakers in several countries consider electric vehicles (EV) as an alternative to fossil-fueled vehicles. In order to allow for the development of EV, the charging infrastructure has to be set up at locations with high charging potential, identified by means of various criteria such as demand density or trip length. Many methodologies for locating charging stations (CS) have been developed in the last few years. First, this paper presents a broad overview of publications in the domain of CS localization. A classification scheme is proposed regarding modeling theory and empirical application; further on, models are analyzed, distinguishing between users, route or destination centrality of the approaches and outcomes. In a second step, studies in the field of *explicit spatial* location planning are reviewed in more detail, that is, in terms of their target criteria and the specialization of underlying analytical processes. One divergence of these approaches lies in the varying level of spatial planning, which could be crucial depending on the planning requirements. It is striking that almost all CS locating concepts are proposed for urban areas. Other constraints, such as the lack of extensive empirical EV traffic data for a better understanding of the driving behavior, are identified. This paper provides an overview of the CS models, a classification approach especially considering the problem's spatial dimension, and derives perspectives for further research.

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Charging station; electric vehicle; GIS; localization; optimization; spatial modeling

1. Introduction

In the face of the still rising greenhouse gas emissions caused by the transport sector in the EU and its large share of total energy consumption (74%) (European Environment Agency, 2017), electric vehicles (EV)¹ are seen as alternative means of transport that might correct this trend (European Commission, 2011, 2016). The introduction of EV has to be complemented by the appropriate infrastructure. The latter requires collective and considerable efforts from countries, regions, and individual municipalities. On the one hand, the development of infrastructure is often emphasized as the main contributing factor for increasing the rate of EV (Beckers, Gizzi, Kreft & Hildebrandt, 2015; National Academy of Sciences, 2015). On the other hand, however, operators of charging station (CS) infrastructure ask for more EV as relevant customers driving the process of expanding the CS network (Nationale Plattform Elektromobilität, 2011). This is the reason why location modeling has a prominent role in this domain, serving as an instrument to forecast the demand that the installation of a CS could induce.

The present development of the number of scientific articles published indicates the high current relevance of this subject. One part of existing publications and research projects looks at the trends and challenges like the insufficient charging infrastructure as a limiting factor for reaching a higher EV market penetration. Another large body of literature presents various strategies for CS location. Therein, the authors address the questions of where the CS within a selected area or along a road network should be placed and how many and which type of CS should be installed.

While some research projects attempt to find the best CS location with an optimization approach using road and driving networks, other studies aim to identify these locations by calculating the spatial distribution of charging demand. Optimization algorithms such as genetic algorithms (Chen, Shi, Chen & Qi, 2015; Hidalgo, Ostendorp & Lienkamp, 2016; Salmon, 2016; Sebastiani, Luders & Fonseca, 2016) or integer programming (Asamer, Reinthaler, Ruthmair, Straub & Puchinger, 2016; Li, Huang & Mason, 2016; Wang, Yuen, Hassan, An & Wu, 2016) are often applied as techniques.

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¹EV is used for fully electric road vehicles in this paper.

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They aim to find the optimal location for CS by minimizing total cost, reducing trip length, or waiting time. However, there are also geospatial (Andrenacci, Ragona & Valenti, 2016; Efthymiou, Antoniou, Tyrinopoylos & Mitsakis, 2012; Frade, Ribeiro, Gonçalves & Antunes, 2011; Namdeo, Tiwary & Dziurla, 2014; Wirges, 2016) and statistical approaches (Brooker & Qin, 2015; Morrissey, Weldon & O'Mahony, 2016; Speidel, Jabeen, Olaru, Harries & Bräunl, 2014) which, for instance, focus on density calculation and clustering methods. Depending on the algorithm and the respective parameters required, the models consider either user or user-group relevant information, and focus on the trajectories of EV or the placement of destinations and their necessity for the users as a basis for the implementation. Existing charging data or tracking profiles of electric or fossil-fueled vehicles as well as statistical, survey, or simulation-based datasets provide the data basis for this locating analysis.

The published literature offers a wide range of CS locating methods (Wirges, 2016); however, never before have these methods been reviewed in such comprehensive manner as in the present review study. One reasonable explanation is that the majority of studies were published recently, following the high amount of attention that the topic has received in the last years. Half of the literature reviewed in this paper was published within the years 2015 to 2016. Previous review work considered a broader, more general view on the placement and technique design of CS (Jing, Yan, Kim, & Sarvi, 2016) or only focused on heuristic approaches, as can be found in Islam (2015) or Shareef et al. (2016). For an optimal placement and sizing of CS, they distinguish between studies with consideration of economic benefits and power grid impacts and on whether the possibility to install slow or fast CS is included or not. Therefore, they divide the techniques according to mathematical algorithms: *Genetic Algorithm*, *Particle Swarm Optimization*, and *Integer Programming*. Further methodical approaches were summarized under the headline of *other techniques*. They were listed with a short description; however, comparative or more detailed classification analyses were not conducted.

This paper presents an extensive overview as well as an in-depth review of the literature dealing with the location of CS for EV. While Shareef et al. (2016) and Islam (2015) focused on the review of the mathematical function of a location model, approaches regarding the *spatial dimension* or *location planning* processes have not been evaluated yet. Hence, this paper is dedicated to this problem's spatial perspective and has three objectives: The first objective is to identify the current state and developments of location modeling for CS infrastructure and to categorize these approaches with respect to their theoretical or empirical focus, modeling orientation (user, route, or destination) and type of outcome (demand-oriented density, partitioning, or network optimization). The second objective is to provide a deeper analysis of suggested and tested models. While a multiplicity of developments are currently prioritized in modeling theory using synthetic data, only a limited number of models were applied in case studies and explicitly considered the spatial dimension. Critical review could contribute to strengthen the theoretical basis of modeling and to identify demands and constraints of

the state-of-the-art literature, e.g. in terms of data input. As a third objective, the key elements for a location model for CS that could be applied and tested in a case study are synthesized. Findings of these works will be considered for a modeling approach in future work.

The following chapters provide a quantitative analysis of research articles to better understand scientific development in CS location modeling and to categorize this research with regard to the application of the derived models on spatial case studies. Subsequently, a new approach for the classification of models was derived from an in-depth analysis of identified papers and is proposed to gain a new perspective and an overview on locating processes. The studies are then discussed in detail with regard to the required input data, theoretical approaches, and maturity (being a test on the basis of a case study represents the highest level of maturity). Finally, the paper points out shortcomings and proposes an approach where recommendations are synthesized from different models. Potentials, on the one hand, and gaps and deficits of selected spatial methodologies and underlying parameters, on the other hand, are identified and serve to provide new perspectives for future work.

2. Overview and literature selection

The number of scientific publications dedicated to the location of CS for EV has grown considerably in recent years. A SCOPUS search for the combination of the terms “electric vehicle” AND “charging station” including related terminology such as “electric car,” “charging infrastructure” and “charging network” in combination (AND) with the term “location” or synonyms such as “placing,” “placement,” “site,” “siting” and “localization” (derived from a search with wildcards for the expressions “plac*,” “sit*,” “local*”) in the title, abstract, and keywords delivered a total of 602 results. The set of publications resulting from the SCOPUS search serves as the basis for the presented review. This set was extended with publications concerning, e.g. EV-related research projects that were identified through the snowball principle, but are not indexed in SCOPUS. Titles and abstracts of a total of 661 publications have been manually reviewed in order to determine which of these publications predominantly deal with models for the optimal location of CS for EV. The collection comprised 284 journal papers (including seven reviews), 323 conference papers (including four conference reviews), 38 books (including two doctoral theses), eight notes, five book chapters, and three reports. The first publication appeared as early as in 1986. It is a research report on the estimated required charging infrastructure in the case of the whole cars fleet in Great Britain being replaced by battery EV (Watson, Gyenes, & Armstrong, 1986). But as indicated in Figure 1, it was only from 2012 onwards that the research topic started to gain considerable attention.

Research related to CS for EV has been conducted in most parts of the world, except in Africa. While the majority of the literature found in SCOPUS has its origin in China and the United States, with 161 and 146 publications,

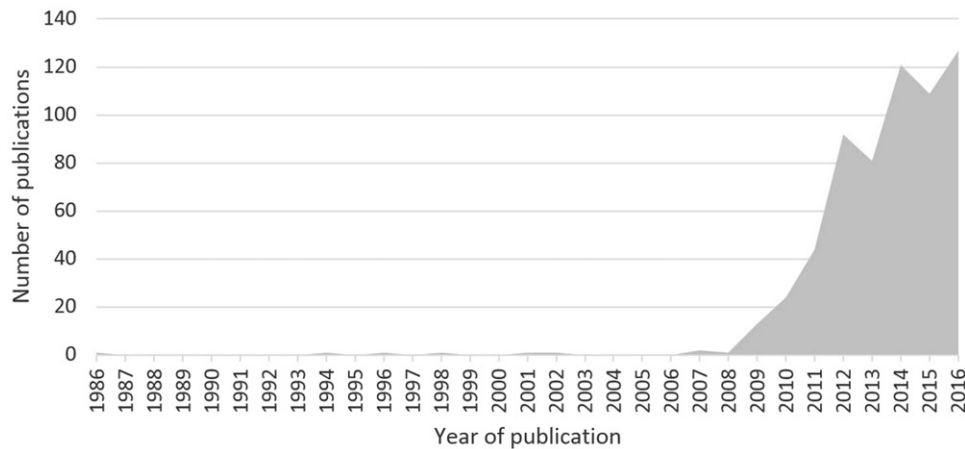


Figure 1. Number of publications in SCOPUS with the terms “electric vehicle,” “charging station,” and “location” and their synonyms (see text) in the title, abstract, and keywords.

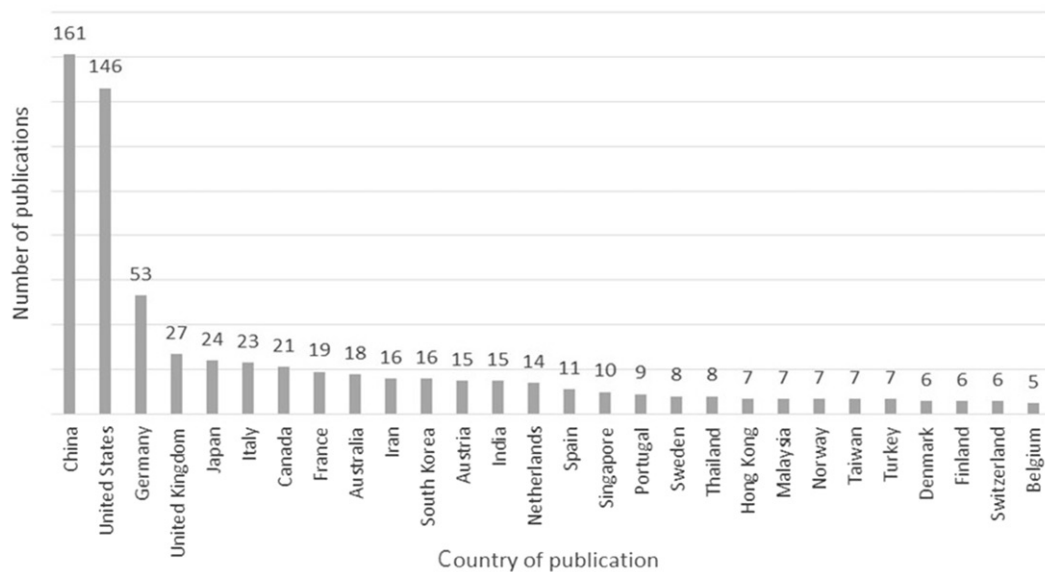


Figure 2. Number of publications of different countries in SCOPUS (origin of the authors).

respectively, the rest of research documents are distributed between other countries worldwide (see Figure 2). The list of countries with research output in the field of EV CS is not limited to high-income countries with a strong automotive industry such as Germany, but extends to countries such as Iran, India, and Malaysia. Research in the latter countries is motivated by governmental targets for the improvement of air quality, e.g. Teheran (Iran) and Beijing (China) show a considerable concentration with respect to these studies.

2.1. Literature without location modeling

From the set of 661 publications found, 498 are related to the topics CS and EV, but do not include a modeling approach for locating CS for EV (see Figure 3). Even though the abstracts, titles, and keywords contain the above-mentioned terms “location,” “electric vehicle,” and “charging station” (or synonyms), they do not aim at finding CS locations. Instead, the discarded publications have their focus on

topics such as the energy supply for CS and their grid integration. These include e.g. the definition of the energy sources for the CS (Ebrahim, 2016; Meyer, Choi, & Wang, 2015; Nahrstedt & Chang, 2016; Sriboon, Sangsritorn, Tuohy, Sharma, & Leeprechanon, 2016), the integration of CS in smart grids (Liu, Kong, Liu, Peng & Wang, 2015; Moghbel, Masoum & Fereidouni, 2015; Nasrallah, Al-Anbagi, & Mouftah, 2016; Van Der Burgt, Sauba, Varvarigos, & Makris, 2015), energy demand management (Castello, LaClair, & Curt Maxey, 2014; Sehar, Pipattanasomporn, & Rahman, n.d.; Zhang, Ai, Gao, & Yan, 2014), power supply quality (Dharmakeerthi & Mithulananthan, 2015; Pinto, Pombo, Calado, & Mariano, 2015; Wang, Cui & Zhu, 2014), the planning of distribution networks (Pazouki, Mohsenzadeh, & Haghifam, 2013; Zhang et al., 2015), and the sizing and siting of CS with relevance for the power grid (Moradi, Abedini, & Hosseini, 2015).

Beside the literature about energy supply and CS grid integration, research articles about the configurations of CS in interaction with EV are discarded. This field includes

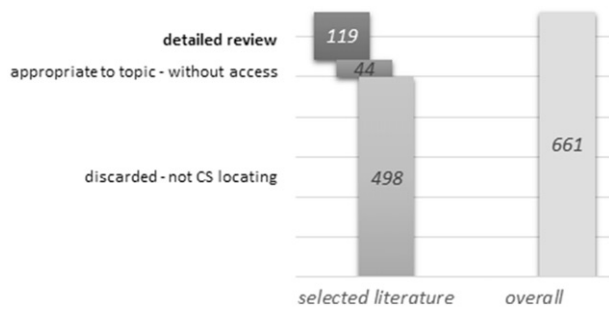


Figure 3. Number of discarded and selected publications for detailed review.

topics such as load modeling (Cai & Shu, 2014; Ghiasnezhad & Filizadeh, 2013; Huang et al., 2016; Tang & Wang, 2013; Yang, Chen, Wen & Chen, 2014), recharge scheduling and dispatch optimization (Amigo & Gagnaire, 2016; Biswas, Gopalakrishnan, & Dutta, 2016; Kulvanitchaiyanunt, Chen, Rosenberger, Sarikprueck, & Lee, 2016), the optimization of queuing and charging (Akbari & Fernando, 2016; Braam, Grosz, Mierau, Kohrs, & Wittwer, 2016; Chen, Huang, Chen, & Jin, 2016; Pinto, Costa, Menasche, & de Amorim, 2016), routing optimization for charging (Arkin, Carmi, Katz, Mitchell, & Segal, 2014; De Weerd, Stein, Gerding, Robu, & Jennings, 2016; Wang et al., 2016; Yang, Yao, Yang, & Zhang, 2016), or the technical development of CS (Bayram, Michailidis, Devetsikiotis, & Granelli, 2013; Stoychev, Tebbe, & Oehm, 2016; Tan, Zhu & Wu, 2016; Waltrich, Hendrix, & Duarte, 2016; Yong, Ramachandramurthy, Tan, & Mithulanathan, 2015), which contributes to the state of knowledge about technical requirements of CS themselves (Ghiasnezhad & Filizadeh, 2013) and also battery systems (Nie & Ghamami, 2013).

Moreover, references found with the above-mentioned keyword search, dealing with economic aspects such as financing schemes (Frade et al., 2011; Sierchula, Bakker, Maat, & Van Wee, 2014; Yang, Long, Li, & Rehman, 2016), business models for CS (Kang, Feinberg, & Papalambros, 2015), market integration (Khoo & Gallagher, 2012; Yu, Li, & Tong, 2014; Yu, Li, & Tong, 2016), pricing strategies for CS (Bernardo, Borrell, & Perdiguer, 2016), and public incentives for the adoption of EV (Mersky, Sprei, Samaras, & Qian, 2016; Silvia & Krause, 2016; Taefi, Kreutzfeldt, Held, & Fink, 2016; Zhang, Xie, Rao, & Liang, 2014) are also not considered for further analysis. Market penetration scenarios for CS and EV (Foley & Gallachóir, 2015; Nemry & Brons, 2010; Plötz, Gnann, Kühn, & Wietschel, 2013) are investigated issues in the EV sector as well, but are also discarded due to the specific review focus without a CS locating approach. Articles about safety aspects (Chung et al., 2013; Gao, Farley, Ginart, & Tse, 2016), authentication and privacy at the charging points (Höfer, Petit, Schmidt, & Kargl, 2013; Li, Dán, & Nahrstedt, 2015; Saini, 2015), social aspects of EV adoption (Elbanhawey, 2015, 2016; Halbey, Kowalewski, & Ziefle, 2015), driver barriers (Haddadian, 2015), and user preferences (Philipsen, Schmidt, & Ziefle, 2015) are also excluded.

The analyses of driving patterns and charging behavior (Azadfar, Sreeram, & Harries, 2015; Brady & O'Mahony, 2016) often include the spatial and temporal distribution of

EV, but do not necessarily have the objective of locating CS. Instead, these analyses evaluate the usage of existing CS (Morrissey et al., 2016; Weldon, Morrissey, Brady & O'Mahony, 2016), driving behavior (Jakobsson, Gnann, Plötz, Sprei, & Karlsson, 2016; Philipsen, Schmidt, van Heek, & Ziefle, 2016), or the charging demand of EV users (Xydas et al., 2016). Ahn et al. (2016) are using their research from a former study about locating CS (Ahn & Yeo, 2015) to analyze driving patterns. While the earlier publication contributes to the review, the second one is discarded. The same applies for a controlled site selection of CS inside of predefined locations like one parking area (Senapati et al., 2016). The rest of publications were kept for detailed revision (see Figure 3). Even if some of the studies contain single elements of those mentioned above, their main issue is the CS location analysis.

From the set of 163 publications considered for detailed review, it was possible to access 119 full texts. Missing full texts correspond to 22 research papers and conference proceedings entirely in Chinese and 22 documents that are not open access and for those, none of the authors has access through institutional subscriptions. Nevertheless, the scientific contribution from the 44 documents without full text access is considered for further analysis based on the information provided in the abstracts or in similar publications from the same authors. Two-thirds of the collective authorship who have published non-accessible literature have published similar articles related to the CS locating topic, which are also included in the review.

2.2. Reviewed literature considering location modeling

Most of the 119 papers are based on studies conducted within the last few years. Three papers are review papers (Islam, 2015; Jing et al., 2016; Shareef et al., 2016). More than half of the reviewed literature was published in 2015 and 2016, the rest between the years 2010 and 2014. This number of publications not only indicates the current investigation demand, but also shows the variety of aspects covered by analyzing methods for the charging location problem, already mentioned by Ahn and Yeo (2015).

These 119 papers provide the basis for a further review on the structural level of the studies as well as on a content level of the models per se (see Figure 4) dealing with the models' orientation and results.

2.2.1. Structural level

From a structural perspective, the reviewed studies are divided into two groups with regard to the model's focus and application, respectively (see Figure 4). The division into theoretical and empirical-based literature is needed in order to identify which models are still at a theoretical construction level and which ones faced a test based on a real-world case study and, thus, show a higher level of implementation orientation and transferability.

The focus of the first class of models lies on the development of the theoretical models. In a smaller part of theoretical approaches, the developed algorithms are tested using

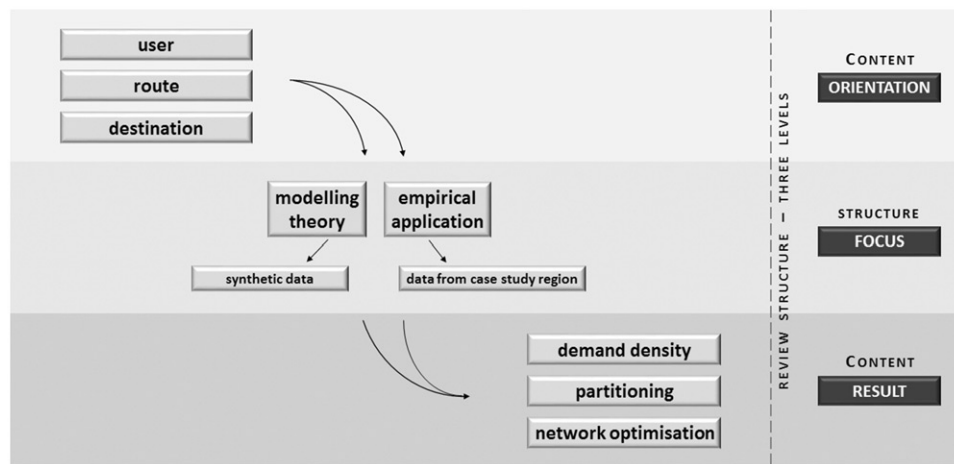


Figure 4. Three levels of categorization on a structural and on a content level.

either synthetic data delivered from a simulation, or secondary data taken from other studies. With the priority on mathematical methodology and only a synthetic test, the first class could be identified as the theoretical class. Studies are assigned to this category even if they contain data of a real investigation region, e.g. the road network or the elevation model, but the significant parameters for EV driving patterns or user behavior are computer-simulated. Hence, they are not real case studies with data specific to the respective region (Bi, Xiao, Viswanathan, & Knoll, 2016; Giménez-Gaydou, Ribeiro, Gutiérrez, & Antunes, 2016; Hiwatari, Iwatsubo, & Ikeya, 2014).

In contrast, the scientific work belonging to the second class can be identified as studies with an empirical focus. Publications in this category apply their models on a specific case study or case study region under the consideration of real data characterizing the traffic, infrastructure, or inhabitants of the investigation area. As these models are already tested in a real, existing environment, they are one step ahead of the studies pertaining to the first class with regard to applicability within a planning process.

55 are studies with a theoretical focus on identifying the optimal CS locations. Almost all of these 55 models are calculated exemplarily on the basis of synthetic data such as simulated data, e.g. traffic data. Three studies only present the mathematical model. Besides these mainly theoretical studies, 61 mainly empirical studies are included in the review selection (see Table 1), as they apply the developed method on a real case study region. Both types of papers consider space, but only the second kind contains a case study with *explicit spatial* context. These studies are defined as approaches with georeferenced data. Synthetic generated data can be used for the theoretical evaluation of methods, but cannot necessarily be considered as a proof of concept for real location planning processes.

2.2.2. Content level

2.2.2.1. Orientation of the models. On a content level, three groups can be identified in an approach to classify the spatial CS locating: the user, route, or destination orientation, which depends on the perspective from which the

requirement for CS is analyzed in a model (see Figure 4). This classification does not only take into account the input data of a model, but input data also determine the analysis process itself. One part of the studies relies on vehicles users' data, quasi on the persons themselves, characterized by their age, social status, daily activities, and their spatial distribution (He, Kuo & Wu, 2016; Hidalgo et al., 2016; Namdeo et al., 2014; Wirges, Linder & Kessler, 2012). The general purpose of this location planning is to identify the number of persons distributed in a predefined area in relation to their lifestyle and residences. To build different EV user groups, the studies often use statistical data such as national statistics on age, gender, or income.

Another group of studies focuses on the destinations. They analyze the type of destination and which type refers to the persons' daily activities. Besides, the dwelling times, the frequency of the visits as well as the visiting numbers of persons are determining the charging requirements close to a destination location (Wagner, Götzinger & Neumann, 2013; Wirges, 2016). Destinations are the target points where people stop and where there is an opportunity for charging. Thus, the user and destination perspectives correlate, but could be divided with regard to the data. While, for instance, Wagner et al. (2013) are using points of interest (POI) in one city as a basis for the destination points, Namdeo et al. (2014) divided the potential EV drivers into three categories generated from a census statistic indicating age group or family size. The first mentioned publication is categorized as destination-oriented, and the second one has a user focus.

As a third perspective, most of the studies set a focus on the travel routes. Especially studies with the main emphasis on modeling theory use network data for simulating the traffic movements on a road network (Dong, Qian, Liu, Zhang, & Zhang, 2016; Lin & Hua, 2015). Wang et al. (2015) define this process as node-based demand assumption and see this as a classical type of location theory. The drivers reside at one node of the traffic network first and need to travel to another node for a certain service or activity. The aim is to optimize the distribution and quantity of CS locations in order to satisfy the demand at the nodes as precisely as possible (Wang et al., 2015; Berman et al., 2010). In these studies, "user behavior" or rather "driver behavior" is often

termed and defines the activity of the driver to decide for a route and how long and where to spend time. Thus, in nearly all cases the drivers' behavior belongs to the route-oriented studies. In the studies with empirical focus, the travel paths are typically evaluated as exact georeferenced paths with tracking information about starting point, end point, and conditions during the trip, e.g. elevation, speed, or driving time.

In conclusion, the reviewed approaches are either based on users, on trajectories or on the destinations visited by EV users. Therefore, the user himself or herself, the driving route or the end point constitute the basis for the modeling process from three different points of view.

2.2.2.2. Result categories. Three categories of modeling results can be identified in the review work: the demand density, partitioning, and network optimization (see Figure 5). Studies such as the one by Frade et al. (2011) analyses the charging requirement for fix determined subareas. They calculate the demand for each area, e.g. in raster format, census blocks or similar divisions, and compare the values of subareas with each other. Hot spots or high-weighted areas are identified as most suitable CS locations. For optimization approaches, the influence of areas in the neighborhood is taken into account (Asamer et al., 2016). The density of the demand can be measured in different units, for example in electric energy (kWh) or the weight depending on the dwelling time of vehicle users.

Another result category describes partitioning. A partition includes density calculation and can identify hot spots, however, without fix determined areas. In the partition concept, the subareas are not divided in specific patterns, but a calculation tool is used to generate partial homogenous areas. Studies with the result type "partitioning" at first calculate the charging demand at individual points. Afterwards, they aggregate the points up to an area of a certain size, using e.g. conversion tools, sum of block statistics (Dong, Liu & Lin, 2014) as well as different clustering methods. Andrenacci et al. (2016), for example, utilize the *k*-means fuzzy clustering method for detecting points with the short-distance of all destinations to possible CS locations.

The last result category is the network optimization result. This kind of result is based on route orientation. This kind of model often uses optimization algorithms with the objective either to minimize total costs, number of CS or transportation costs, or to maximize the coverage demand named by several of the studies (Islam, 2015). Individual placements for CS within the road and travel network result out of these flow-refueling location problem alike studies (Kuby & Lim, 2005, 2007). This type can be combined with the other result categories demand density and cluster, as the single points can be aggregated to demand areas like in the modeling process of Dong et al. (2014).

3. Analysis of explicit spatial location methods

For a more detailed literature review, 61 publications are selected in which real case studies with *explicit spatial reference* show the models' applicability on the basis of real-

world data (see Section 2.2). What all studies have in common is the objective of identifying specific and georeferenced points or hot spot areas where the CS should be implemented in contrast to studies only with the result of a number of CS within a whole district of a municipality. These ones are chosen because the mere number of CS for the whole municipality or at a district level is insufficient for planning purposes and project realization. Instead, the review follows the idea that only a locating approach can support the stakeholders in the decision-making process.

For the detailed review, the selected modeling approaches are classified with regard to data sources (Section 3.1), relevant parameters (Section 3.2), optimization algorithm and specialization (Section 3.3) as well as the case study area (Section 3.5).

3.1. Data sources for different modeling approaches and their constraints

Relevant data are determined by selecting the processing method for identifying potential CS locations. To obtain these parameters, the following data sources can be taken into account:

1. Statistics, e.g. census data
2. Travel data of EV and measured CS data
3. Travel data of fossil-fueled vehicles
4. Surveys (questionnaire)
5. Simulation or test data

Statistics such as census data are a frequently used source in user-oriented models. As example, the model of Wirges et al. (2012) is based on a residential population and the number of commuters for evaluating the demand density. Efthymiou et al. (2012) take the level of income into account as a criterion for the willingness to buy an EV and estimate the spatial distribution of EV in a city. Namdeo, Tiwary & Dziurla (2014) distinguish between different user groups, utilizing sociodemographic statistics and using family sizes, ownership and number of vehicles, age groups, profession and income as well as commuting patterns. The study also combines the personal user characteristics with travel patterns. The statistical data characterize different user groups and deliver a fast overview with a relatively small dataset. With specific characteristics of a region, the electric consumption and thus the demand potential for CS can be assessed. However, as is the nature of statistics, the equal characteristic value for a user group (one statistic value for a whole district) constrains the models, as they assume the same conditions without a detailed spatial distinction (e.g. real number of residents per house). The same applies to the forecast of market penetration rates for EV and CS. The extent to which the number of EV will increase in the coming years can hardly be estimated (Wirges, 2016), which additionally renders the predicted number of EV in relation to the total quantity of vehicles or the number of residents all but vague. Collecting data for identifying areas with a high demand density for implementing CS, after all, always

Table 1. Target criteria for locating CS in each reviewed study based on a structured selection of relevant literature considering only papers based on empirical data (as described in Section 2.2).

Reviewed literature	User/destination orientation				Route orientation				EV range (SOC)	CS range	Finance cost
	Demand density	Distance user—CS	Distance destinat.—CS	Dwell time	Trip length	Traffic density	Travel time	Queuing			
Ahn and Yeo (2015)					x	x					x
Andrenacci et al. (2016)						x				x	
Asamer et al. (2016)						x				x	
Baouche et al. (2014)					x						x
Böhm (2013)					x		x				x
Brooker and Qin (2015)	x	x		x							
Cai and Shu (2014)	x			x							
Cavadas et al. (2014)	x		x	x							x
Chung and Kwon (2015)						x					
Cruz-Zambrano, Corchero, Igualada-Gonzalez, and Bernardo (2013)	x										x
Dong et al. (2014)				x	x					x	x
Efthymiou et al. (2012)	x		x								
Eisel, Schmidt, and Kolbe (2014)				x ^a	x	x					
Frade et al. (2011)	x	x	x								
Gao and Lu (2015)	x				x ^b						
Gkatzoflias, Drossinos, and Zambelli (2016)	x		x								
Guo and Zhao (2015) ^c											
Han et al. (2016)						x	x	x	x		x
He et al. (2016)	x	x									
Hidalgo et al. (2016)				a		x			x		x
Huang et al. (2016)			x			x				x	x
Islam, Shareef, and Mohamed (2016)						x					x
Islam, Mohamed and Shareef (2015)	x										x
Jia, Hu, Liang, Lang, and Song (2014)	x										x
Kameda and Mukai (2011)					x		x				
Kang, Feinberg, and Papalambros (2016)							x	x	x		x
Li et al. (2015)							x	x			
Li et al. (2016)					x						
Li and Huang (2014)							x	x			
Liu, Zhang, Ji, and Li (2012)						x					x
Luo, Huang, and Gupta (2015) and Luo, Huang, and Gupta (2016)	x										d
Namdeo et al. (2014)	x										
Rajabi-Ghahnavieh and Sadeghi-Barzani (2016)	x										
Rastegarfar, Kashanizadeh, Vakilian, and Barband (2013)											x
Ravlić, Erdelić, and Carić (2016)			x		x ^d						x
Sadeghi-Barzani, Rajabi-Ghahnavieh, and Kazemi-Karegar (2014)	x									x	
Salmon (2016)						x			x		x
Sathaye and Kelley (2013)					x	x					x
Sebastiani et al. (2016)					x						x
Shahraki, Cai, Turkay, and Xu (2015)					x				x		
Sheppard, Gopal, Harris, and Jacobson (2016)		x		x							
Speidel et al. (2014)		x									
Spieker et al. (2016)	x		x			x					
Tu et al. (2016)	x			x	x			x	x		
Viswanathan et al. (2016)	x				x				x		
Wagner, Brandt and Neumann (2014)	x ^f		x							x	
Wagner et al. (2013)	x		x	*						x	
Wirges (2016)	x	x								x	
Wirges et al. (2012)	x	x									x
Wood, Neubauer, and Burton (2015)				*	x				x	x	
Xiang et al. (2015)						x		x			x
Xiang et al. (2016)						x		x			x
Yağcıtekin, Uzunoğlu, and Karakaş (2016)	x		x					x			x
Yi and Bauer (2014) and Yi and Bauer (2016)					x					x	
Zhang et al. (2015)					x			x	x		x
Zhang, Qi, Jiang, and Lei (2016)	x							x			x
Zhang, Hu, Xu, and Song (2016)	x							x		x	x
Zhao and Li (2016)											
Zhu, Gao, Zheng, and Du (2016)	x										x

^aThese studies assume that the dwell time is sufficient for loading (Hidalgo et al., 2016; Sebastian Wagner et al., 2013; Wood et al., 2015) or that a long dwell time for charging is given during day-time (working time) and night-time (resting at home) (Frade et al., 2011).

^bAssumption of a CS density on a street grid of a station every 30 km.

^cEvaluation of pre-given CS regarding sustainability criteria (social, economic, environmental).

^dOnly in theory model.

^eIf the EV range (SOC) is over 20%, an extra modeling for minimal charging detours is done.

^fDemand density calculation in terms of weight of a POI.

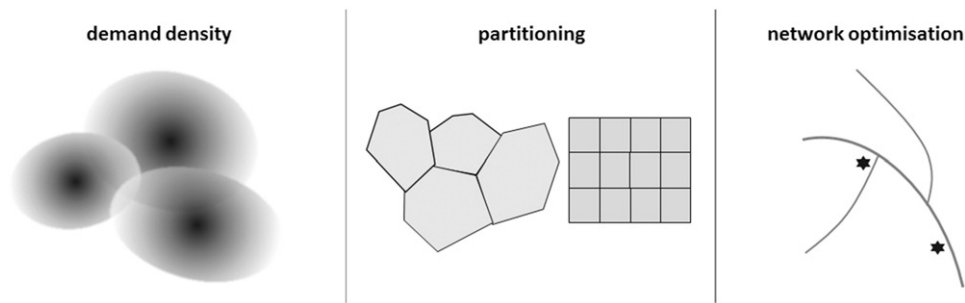


Figure 5. Scheme of result categories for CS location modeling.

constitutes an inexact assessment based on several assumptions. However, most of the studies apply highly aggregated statistical data with a low spatial resolution (e.g. criterion demand density in Table 1, 26 out of 61 studies) (Spieker, Hagg, Gaier, Meilinger & Asteroth, 2016; Viswanathan et al., 2016).

CS locating methods which do not consider statistical data use existing travel data, mostly collected from GPS-tracked, conventional vehicles. These approaches assume that EV driving behavior complies with the use of conventional vehicles. In their modeling process, Dong, Liu & Lin (2014), for instance, take into account real travel data of fossil-fueled cars and describe the consistent condition as “[...] motorists’ travel behaviour remains unchanged when switching to BEV technologies.” The remaining user patterns mean a complete transition of travel condition and, actually a penetration rate of 100%. By investigating not the demand density but the route characteristics, specifications about the vehicles’ trajectories such as the travel distance and battery consumption are essential. This is applied by the models of Andrenacci et al. (2016), Asamer et al. (2016) or Dong et al. (2014). Travel data deliver the exact location of a trajectory’s starting and end point, temporal transparency about departing and arriving time and the dwelling time at a destination (time between stop at endpoint and start at starting point). The model implementation hereby is based on single agents acting individually in space and time. In contrast to statistical data modeling, agent-based models define exact geographical allocations for CS depending on the tracked user behavior. However, the knowledge about all agents’ activities within the research area is restricted by the knowledge about the single agents and must be simulated for all vehicle drivers. The assumption of equal activities and travel routing of non-observed drivers therefore is a constraining factor (Frade et al., 2011).

Other than dealing with travel data from conventional vehicles, travel data from existing EV tracking can be used for location modeling. In this case, there is no need to assume that the drivers’ patterns may change when switching from combustion to EVs. These data are already analyzable under real conditions (Ahn & Yeo, 2015; Han, Ahn, Park & Yeo, 2016). Also the occupancy of existing CS has already been evaluated (Wagner et al., 2013). However, as until today the share of EV is still on a low level (e.g. as measured by the EV share of new sales in 2015 (European Environment Agency, 2016), which is under 3% in almost all EU countries), the validity of this data for a whole set of

travel paths from a potential EV fleet is not considered. Hereby, the agent-based method also serves for the modeling on the basis of this data source. Exceptions within this type of data source are toll gate detection and counts, which deliver traffic density information for specific road intersections and can thereby estimate the spatial and temporal traffic distribution in a specific research area (Yagcitekin & Uzunoglu, 2016). Simulation or test data are also used in the CS location models, particularly in the studies with modeling theory focus. These data are mostly travel data of EV or conventional vehicles.

Surveys are another possibility for gathering information about the demand for CS depending on the questioned persons and their travel activities. In contrast to the GPS tracking studies, the exact travel routes cannot be measured by questioning the drivers. Driving behavior, origin and destination as well as temporal information are more general as the respondents name time and places, e.g. with the help of a diary (Zhang, Shaffer, Brown, & Scott Samuelsen, 2015). Furthermore, questioning might complement the research for CS location analysis with meanings and expectations of potential drivers (Wirges, 2016). In this respect, it constitutes a shortcoming that the surveyors derive a meaning from a hypothetical situation which they can only imagine. It is not known in which way the expected behavior changes or persists.

3.2. Target criteria for locating CS

Although all analyzed studies have different approaches, they aim to achieve suitable locations for CS. The criteria by which the models determine the CS locations are summarized in Table 1. In general, the models somehow try to cover or satisfy the charging demand with a minimal number of CS with budget reasons on the one side and the goal of achieving EV range on the other side. This may be reached by different criteria according to the three identified model orientations: First, in the user and destination-oriented models, *demand density* is defined as one of the target criteria, specifically requiring the spatial and quantitative distribution of demand indicated by the origin or destination of trips or the residence of users (As to the similar requirements and targets, the criteria are identical for destination and user-oriented models.) Another orientation is the distance between the origin (*distance user—CS*) and the next CS or the destination of a trip and the next CS (*distance destination—CS*). Distance can be measured in spatial length or in time. *Dwelling*

time, which is also the parking time for EV, is another criterion for identifying the charging demand at different locations.

Second, route orientation is expressed in models by *trip length* (total driving distance from origin of a trip to CS), *traffic density* (density of vehicles per route section) and *travel time*. For the travel time duration of a trip, in some cases also the waiting time for charging and queuing as well as the travel or walking time between CS and destination, but not the charging time itself, are included.

In contrast to demand density taking into account static user or destination information, the traffic density is determined by the density of trips in a defined spatial area or road section. In detail, some models also consider the *queuing* and optimization of queuing in front of a CS. Also the state of charge (*EV range SOC*) is used as a factor for identifying suitable locations for CS.

As third and overarching criteria, *financial costs* consider e.g. installation and operation costs of CS. Besides, the target criterion *CS range* can be defined as the spatial coverage of a CS in all directions, e.g. based on the street graph or radius.

The target criteria depend on the model, which is either demand/user-oriented or route-oriented. The most common criteria for CS locating are demand density with a number of 26 and financial costs with 28 out of 61 reviewed cases. For route orientation and therefore often used travel information, the trip length or traffic density are important factors in 18 and 15 cases, respectively. While some studies focus only on one criterion, the majority of studies has two or three criteria on which they base their models. Dong et al. (2014) or Hidalgo et al. (2016), for example, adopt the approach not only to minimize the installation costs, but also to minimize the amount of failed trips. Only six studies present models that are user/destination-oriented and traffic-oriented (Dong et al., 2014; Eisel et al., 2014; Huang et al., 2016; Spieker et al., 2016; Tu et al., 2016; Viswanathan et al., 2016).

Furthermore, the target criteria in the analyzed studies are diversely characterized. For example, financial cost oriented concepts consider, beside CS installation (Dong et al., 2014) or installation and operation costs (Liu, Zhang, Ji, & Li, 2012; Zhang, Qi, Jiang, & Lei, 2016), delay time costs, access costs (Yongjun Ahn & Yeo, 2015), EV costs (Namwoo Kang, Feinberg, & Papalambros, 2016), or opportunity costs (Salmon, 2016). Grid-related costs are an extension to cost orientation as well (Sadeghi-Barzani, Rajabi-Ghahnavieh, & Kazemi-Karegar, 2014; Xiang et al., 2016; Yagcitekkin & Uzunoglu, 2016). As to trip length, the aim is partly to minimize the trip length or time by installing the CS at optimal places. It is also possible to aim at a maximal trip length or a maximal amount of vehicle miles traveled being electrified (Shahraki, Cai, Turkay, & Xu, 2015), to minimize users' losses on the way to the CS (Ge, Feng & Liu, 2011), to minimize the energy consumption (Yi & Bauer, 2016) or to calculate on the basis of trip length the energy consumption delivering demand density (Viswanathan et al., 2016). The last case shows that in most models more than one criterion is needed.

In addition, the distance between the end of a route and the destination consequently is the walking distance (if there

is no public transport following) which the vehicle user has to walk from the parking or charging place to the destination itself. The limits are not based on studies on acceptable walking distances, but are always assumed in the form of walking length or time (Cavadas, Correia, & Gouveia, 2014; Frade et al., 2011). The same applies for penetration rates which are assumed on a more or less hypothetical basis or with regard to political targets and with approximate values, as forecasts in this early stage of market penetration are vague (e.g. Jia et al., 2014; Wirges et al., 2012).

3.3. Optimization algorithms and specialization

Islam (2015) and Shareef, Islam & Mohamed (2016) reviewed CS locating studies with regard to the optimization algorithms and categorized the model algorithms into Generic Algorithms (GA), Particle Swarm Optimization (PSO), Integer Programming (IP), and other techniques. The therein reviewed approaches utilize the spatial dimension in several steps of the modeling process, but not every study incorporates the spatial dimension in each and every process step. Each of the models has spatial data as input layer, such as geotagged trajectories of conventional cars or EV. Even the statistical values belong to a specific spatial area for which the demand of EV or CS is calculated. The distinction between the models can be seen in the analysis steps as the studies use either: (1) only optimization algorithms without geospatial analysis or (2) the analysis process is a combination of an optimization algorithm and a geospatial analysis. The third option (3) is an entirely geospatial analysis without an optimization algorithm (see Figure 6).

There is a multitude of examples that can be classified into the first group of approaches. For instance, the studies of Jia et al. (2014) and S. Li et al. (2016) rely on georeferenced data as input and present the results in form of vector maps with the punctual location of CS, but they do not apply any geospatial analysis for finding these locations. Instead, the authors of these studies propose the CS siting problem as mixed-integer linear program (MILP) with cost minimization functions as target equations, which were solved with the IBM ILOG CPLEX solver or with GA. Chen, Shi, Chen & Qi (2015); Hidalgo, Ostendorp & Lienkamp (2016); Salmon (2016) and Sebastiani, Luders & Fonseca (2016) also use GA for specifying the circumstances and parameters in general and identify the optimal CS locations with the input of e.g. network and trip length or the battery SOC. Other studies such as those by Islam, Shareef & Mohamed (2016), Li & Huang (2014) and Liu et al. (2012) follow the same approach, but use another type of mathematical description of the problem and other solvers.

Studies like the ones by Asamer et al. (2016) or Tu et al. (2016) belong to the second group of modeling type. These authors divided the analysis in two steps. First, in the case of Asamer et al. (2016), hexagons that cover the maximal number of taxis' trips in the city of Vienna, Austria are identified. The mathematical problem of finding the hexagons is described in form of an MILP that was solved with the IBM ILOG CPLEX solver. Second, a geographic

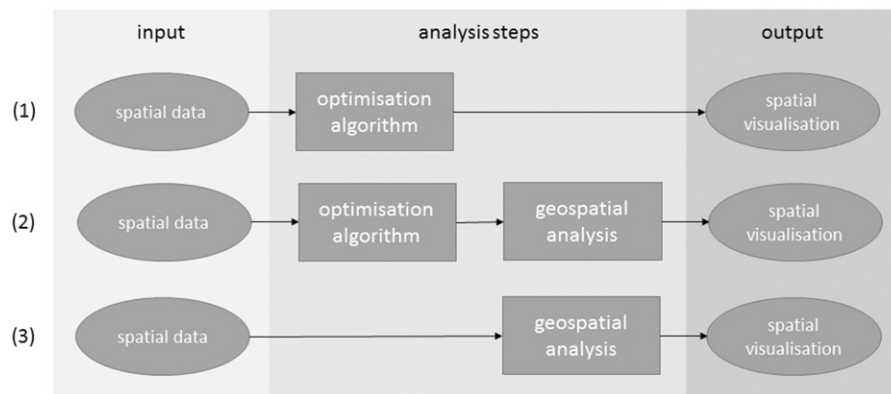


Figure 6. Classification of the modeling process.

information system (GIS) is used to determine the centroids of the hexagons and to calculate isochrones of five or eight minutes for the EV to reach a CS. The results of the latter two calculations serve to define the potential CS placements. The centroids of the hexagons are calculated and represent the punctual locations for the CS. Asamer et al. (2016) chose the hexagons as suitable potential areas for CS with a diameter of 1 km or rather a travel time of approximately 4 min to get a similar distance from the centroid point to all neighbored hexagons without overlapping. Thus, the interaction with the neighbored area can be considered. Hence, the optimization tool is taken into account only with spatial input data but without geospatial methods, while the localization and neighboring relations are calculated in GIS. As another example, Wagner et al. (2014) developed a decision support system for CS location planning by using the located point of interests in the city of Amsterdam, Netherlands, and already existing CS for calculating the highest CS utilization. First, a regression model is used for determining influencing factors such as sociodemographic factors or the type of point of interest. Second, in a geospatial approach, the inner city area was divided in sections of 100 m length to distribute the CS demand due to the regression results. A heat map and exact placements of new CS are proposed.

A few studies are categorized in the third group of complete geospatial CS location analysis. They do not only use geospatial data and visualize results in form of a map, but their complete CS location model is based on geospatial analysis. Namdeo et al. (2014) offer such a modeling approach. With a multidimensional spatial analysis, the study includes socio-economic factors and combines commuting, vehicle, and household numbers by using different GIS layers. The values vary due to the divergence of the administration units in the Tyne and Wear region, United Kingdom. A spatial plot shows the outputs from weighted overlay statistics for public charging locations. Efthymiou et al. (2012), Gkatzoflias et al. (2016) and Wirges (2016) use similar approaches by transferring statistical data into the spatial dimension for identifying areas with high charging demand in GIS. Andrenacci et al. (2016) adopted a distance-based approach by calculating the charging demand for EV from tracking data in the case study of Rome, Italy. With a k-mean clustering approach, they minimize the sum of distances between potential charging points and areas with high

charging demand. Hence, the results are visualized as hot spot areas, exact georeferenced potential CS or for different sections, derived from a partitioning (see also Section 2.2).

3.4. Case study regions

Case studies for applying CS location models have been conducted almost all over the world (see Figure 7), except in Africa. While a few studies use a whole country as testing area and allocate CS all over the country (Chung & Kwon, 2015), another part looks at research areas on the county level (Namdeo et al., 2014; Wirges, 2016; Zhang et al., 2015). But in most of the cases, cities, districts of cities or cities and the metropolitan areas are used as case study areas (Andrenacci et al., 2016; Frade et al., 2011; Kameda & Mukai, 2011; Sadeghi-Barzani et al., 2014; Wagner et al., 2014; Wood et al., 2015; Zhao, Zhang, Wei, & Zhang, 2015). Only a few studies apply the models in a small area below an administration level as they observe only a section of a district or city area (Böhm, 2013; YağCitekin, Uzunoglu, & Karakaş, 2016). In Iran and China, studies are often motivated by governmental targets for the improvement of air quality, for example in cities like Teheran (Iran) (Rajabi-Ghahnavieh & Sadeghi-Barzani, 2016) or Beijing (China) (Gao & Lu, 2015).

Most of the models are tested in urban areas, in cities or within a metropolitan area of a city, because of the strengths of electric mobility that are seen in this environment. High population density is characteristic for cities and their service areas. Generally spoken, this means that a higher demand density of CS for EV exists in comparison to rural areas (assuming a similar penetration rate of EV). Moreover, data of tracked trips are particularly available in cities, e.g. Shenzhen (Jia et al., 2014; Shahraki et al., 2015; Tu et al., 2016). But the challenge for covering the demand is considerably higher in rural than in urban areas, as the population is more scattered and the driving ranges between the destinations are longer (infrastructure facilities partly allocated with regard to population density). Especially due to range anxiety and the improvement of recharging efficiency, as some of the location studies emphasized (Ahn & Yeo, 2015; Wang et al., 2016), CS locating problems in rural areas should be solved so that EV can be used for daily journeys. However, several models seem to be transferable from urban

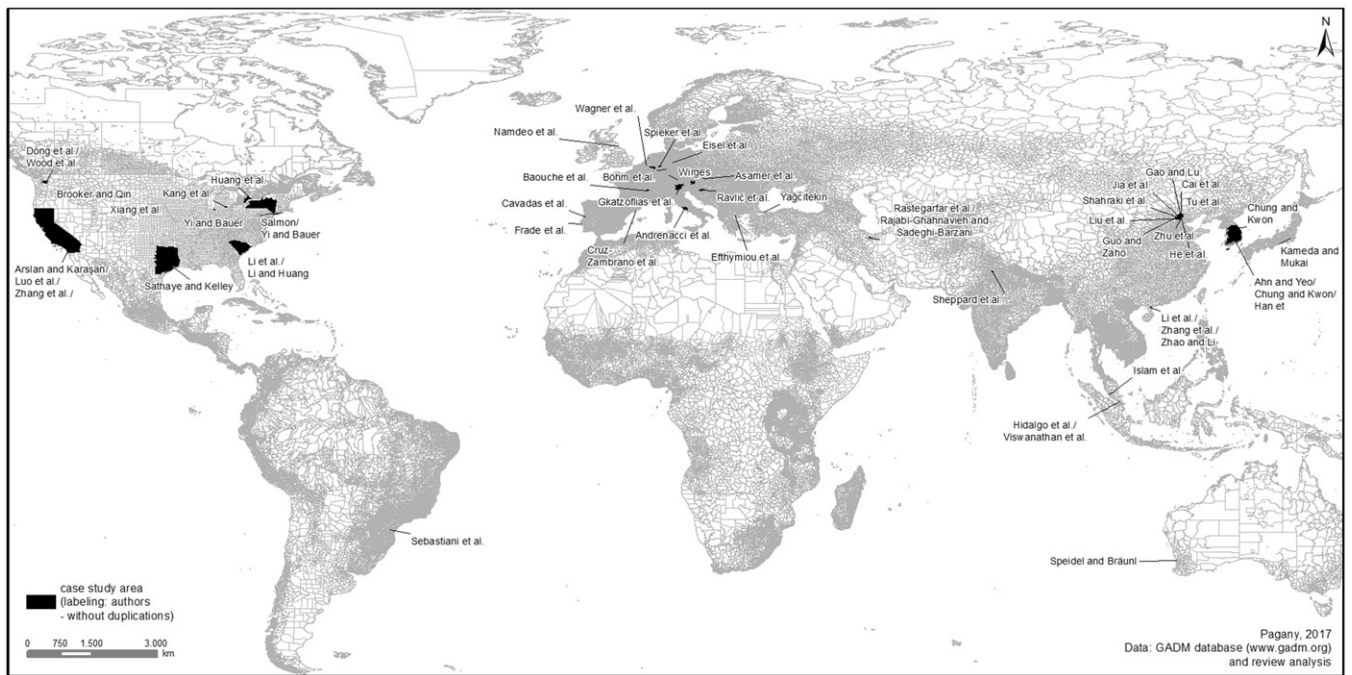


Figure 7. s Case study regions (with authors' label) of reviewed literature.

to rural case studies with regard to the applied methodological approach and supposed that relevant data are also available there, but the proof is missing.

4. Conclusion and perspectives

The present paper provides a comprehensive overview of the state-of-the-art literature for CS location analysis. It categorizes the literature with view to the modeling approach and provides specific information, especially with regard to the criteria applied for identifying CS locations. It shows that location planning models for CS use a wide variety of theoretical approaches and attempts to localize CS with different regional emphasis. The diversity of reviewed models ranges from statistical distribution analysis on a country level to the concrete localization of demand zones or placements on a street level.

Nonetheless, only a restricted number of studies with an *explicit spatial* focus were identified and especially the analysis of rural areas is a gap. Only 119 out of 661 studies have a spatial CS locating approach and only 61 are empirically tested. Geospatial analyses of the EV transport and CS location situation were only found in a limited number of studies. Even in detailed analyses of locations based on neighborhood relations, space is not taken into account or is reduced to an analysis of weighted POIs. This ignores the established location theories applied in other disciplines such as retail or tourism which found that neighborhood as well as distance are essential in location modeling (Krugman, 1997). In the same way, an analysis of the choice of covering distance for CS for road EV in the scope of the theory framework proposed by Berman et al. (2010) is missing in the reviewed literature and has only been part of studies concerning “non-road” vehicles (Fekete, 2017; Fekete, Lim, Martin, Kuhn, & Wright, 2016). The main focus of the

reviewed studies is on urban agglomerations and metropolitan areas. Special attention is given to cities with air quality problems. While in urban areas, traffic and environmental problems seem to be the main drivers for studies, the problem of future individual mobility in rural areas in a post-carbon society has so far not been addressed in CS locating studies. A gap is seen in research on rural areas for defining useful approaches for CS locating there, particularly in terms of the divergent conditions between rural and urban regions.

The selection of the approach and the testing area in the analyzed studies seems to be driven by data and data availability and a lack of data is a main obstacle. There is a high concentration of studies in areas where data sets are available (taxi data, fleet tracking). A substantial number of studies are restricted in their methodology by lack of wide area data. This may also explain the low number of rural studies. Almost the half of studies reviewed in detail are based on demand density, which means that predominantly statistical data of population or car density on a regional level were used. Studies with a higher spatial resolution are mainly based on tracked travel data and optimize on the basis of trip length and traffic density. Only a few studies link origin and destination of travels with the existing POI in the neighborhood or the documented purpose of a trip and so take into account neighborhood aspects from a CS's perspective. A better connection of data sources could provide an improved data basis for future studies. This could contribute to coupling user and destination-oriented models with traffic-oriented models which are identified as the main CS location modeling approaches. All studies show that the access of data is limited, either to the spatial extent or the type of traffic. Studies use small-scaled tracked travel data, either from taxis, buses, or fleets or general street traffic monitoring. These data are often projected to the total research area, upscaled to the whole traffic situation or transferred from specific traffic to individual

transport, but extensive empirical traffic data sets for testing real behavior are missing.

The categorization and quantification of the criteria applied in studies not only shows that studies depend on data availability, but indicates some trends and new approaches for modeling. Beside the most important valued criteria like the cost factor as hereby most founded criterion (considered in nearly every second study), CS charging speed is a factor considered in some studies and would be worth reviewing in more detail. Temporal aspects, the user types and their behavior are not explicitly taken into account in most studies, but could have a strong influence in CS location modeling. Due to the restricted set of parameters used in analyzed studies, for future research more integrated models should be considered to take further relevant parameters into account.

With regard to the identified criteria for modeling the CS location analysis on the basis of the existing literature, it can be concluded that new and more integrated approaches should be developed in a next stage. Conceivable would be the minimization of the walking distances between the CS and the destination locations under consideration of the neighborhood aspect and, in addition to that, the weight of the user behavior, which means the daily activities of persons, their frequency of travel to, travel distance to and dwell times at the destinations. Such an approach could, in a holistic way, connect the route with the user and destination-oriented dimension.

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